

2. NEWTON'S METHOD FOR SYSTEMS OF EQNS

- ANALOGUE OF $f'(x)$ HERE IS JACOBIAN MATRIX

J OF FIRST DERIVATIVES WITH ELEMENTS J_{ij}

$$J_{ij} = \frac{\partial f_i}{\partial x_j} \quad (3)$$

- CURRENT EXAMPLE

$$f_1(x, y) = \sin(xy) - \frac{1}{2}$$

$$f_2(x, y) = y^2 - 6x - 2$$

$$\underline{J} = \begin{bmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y \\ \partial f_2 / \partial x & \partial f_2 / \partial y \end{bmatrix}$$

$$= \begin{bmatrix} y \cos(xy) & x \cos(xy) \\ -6 & 2y \end{bmatrix}$$

- DERIVE MULTI-DIMENSIONAL NEWTON ITERATION, USING MULTIVARIATE TAYLOR SERIES EXPANSION, PARALLEL 1-D CASE

$$\underline{0} = \underline{f}(\underline{x}^*) = \underline{f}(\underline{x}^{(n)}) + \underline{J}[\underline{x}^{(n)}] \cdot \underline{(\underline{x}^* - \underline{x}^{(n)})} + O((\underline{x}^* - \underline{x}^{(n)})^2)$$

↪ column vector

NOTATION: $\underline{J}[\underline{x}^{(n)}]$ → EVALUATE J_{ij} AT $\underline{x} = \underline{x}^{(n)}$

- DROP HIGHER ORDER TERMS, REPLACE $\underline{x}^*$ WITH $\underline{x}^{(n+1)}$

- DEFINE $\underline{\delta x}^{(n)}$

$$\underline{\delta x}^{(n)} \equiv -(\underline{x}^{(n+1)} - \underline{x}^{(n)})$$

D-DIMENSIONAL NEWTON ITERATION IS

$$\underline{x}^{(n+1)} = \underline{x}^{(n)} - \underline{\delta x}^{(n)} \quad (4)$$

WHERE UPDATE VECTOR $\underline{\delta x}^{(n)}$ SATISFIES DND LINEAR SYSTEM

$$\underline{J}[\underline{x}^{(n)}] \underline{\delta x}^{(n)} = -\underline{f}(\underline{x}^{(n)}) \quad (5)$$

- AGAIN, NOTE THAT $\underline{J}[\underline{x}^{(n)}]$ HAS ELEMENTS

$$J_{ij} [x^{(n)}] = \left. \frac{\partial f_i}{\partial x_j} \right|_{y=x^{(n)}} \quad (6)$$

- AT EACH NEWTON STEP LINEAR SYSTEM (5) CAN BE SOLVED IN MATLAB USING MATRIX-VECTOR LEFT DIVISION

GENERAL STRUCTURE OF A MULTI-D NEWTON SOLVER

x : SOLUTION VECTOR

res : RESIDUAL VECTOR

J : JACOBIAN MATRIX

dx : UPDATE VECTOR

$$\|v\| = \left(\sum_{i=1}^n v_i^2 \right)^{\frac{1}{2}}$$

I.E. RMS VALUE

$$x = x^{(0)}$$

while $\|dx\|_2 > \epsilon \rightarrow$ PRE-CHOSEN CONVERGENCE CRITERION

for $i = 1:neg$ — # OF EQUATIONS, D

$$res(i) = f_i(x)$$

for $j = 1:neg$

$$J(i,j) = [\partial f_i / \partial x_j](x)$$

end

end

$$dx = J \backslash res$$

$$x = x - dx$$

end

- HERE, AND FOR 1-D CASE (HOMEWORK!) SHOULD MODIFY THE ABOVE TO LIMIT THE NUMBER OF ITERATIONS (LESS THAN OR EQUAL TO SOME SPECIFIED NUMBER) TO GUARD AGAINST NON-CONVERGENCE INTRODUCTION TO FINITE DIFFERENCE APPROXIMATION (FDA)

1. BASIC IDEA VIA SIMPLE EXAMPLE

PARTICLE MOTION IN 1-D: POSITION/DISPLACEMENT $x(t)$

- SUPPOSE WE ARE GIVEN VELOCITY

$$v(t) \equiv \frac{dx}{dt}$$

SO, FOR ANY t WE CAN EVALUATE NUMERICALLY $v(t)$

- CAN APPROXIMATE VELOCITY USING

$$v(t) \approx \frac{dx}{dt} \approx \frac{x(t+\Delta t) - x(t)}{\Delta t}$$

Δt IS SOME INCREMENT OF TIME: AS $\Delta t \rightarrow 0$ THE APPROXIMATION BECOMES INCREASINGLY ACCURATE

- SOLVE FOR $x(t+\Delta t)$

$$x(t + \Delta t) \approx x(t) + \Delta t v(t) \quad (*)$$

(KNOW THIS)

- ONCE WE KNOW, SAY, $x(0)$ CAN DETERMINE $x(\Delta t)$, $x(2\Delta t)$, ...

$$x(\Delta t) = x(0) + \Delta t v(0)$$

$$x(2\Delta t) = x(\Delta t) + \Delta t v(\Delta t)$$

$$x(3\Delta t) = x(2\Delta t) + \Delta t v(2\Delta t)$$

⋮

$$x((n+1)\Delta t) = x(n\Delta t) + \Delta t v(n\Delta t)$$

⋮

ILLUSTRATES THREE KEY STEPS IN SOLVING ODES
($\oplus$ PNEE) USING FDA

- (. DOMAIN): REPLACE CONTINUOUS INDEPENDENT VAR (t) WITH DISCRETE VALUES $0, \Delta t, 2\Delta t, \dots$